Emma G. Cunningham

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Education

2027*	**Ph.D.** in Psychology, University of Wisconsin-Madison - Dissertation Title: *"Deconstructing Planning: An Examination of Planning Measurement Through Novel Game-Based Paradigms"*
2023	**M.S.** in Psychology, University of Wisconsin-Madison
2021	**B.Sc.** in Cognitive Science–Emphasis in HCI & AI, University of California-Santa Cruz

Teaching

University of Wisconsin-Madison

2026	**Instructor of record:** Introduction to Psychology - *online* - 90 students; introductory level undergraduate course
2025	**Instructor of record:** Games, Technology & Psychology - 24 students; capstone/upper-level undergraduate course
2025	**Teaching Assistant:** Introduction to Psychology - *online* - 90 students; introductory level undergraduate course
2024	**Teaching Assistant:** Introduction to Psychology - *online* - 100 students; introductory level undergraduate course
2024	**Teaching Assistant:** Research Methods in Psychology - 115 students; upper-level undergraduate course
2023	**Teaching Assistant:** Introduction to Psychology - 200 students; introductory level undergraduate course
2022-2023	**Teaching Assistant:** Design and Analysis of Psychological Experiments I & II - 25 students; graduate level statistics course

*Anticipated degree completion in May 2027.

University of California-Santa Cruz

2019 **Peer Tutor:** Sensation & Perception
- 30 students; upper level undergraduate course

Academic Mentorship

Undergraduate Theses and Mentored Independent Projects

2026-2027 **Addison Peters** *(UW-Madison)*, Individual Research Project to be presented at the Undergraduate Research Poster Symposium

2026-2027 **Conlan Yourell** *(UW-Madison)*, Senior thesis project to be presented at the Undergraduate Research Poster Symposium & The Student Thesis and Research Showcase

2026-2027 **Chaoran Li** *(UW-Madison)*, Senior thesis project to be presented at the Undergraduate Research Poster Symposium & The Student Thesis and Research Showcase

2026-2027 **Alexander McGinty** *(UW-Madison)*, Individual Research Project to be presented at the Undergraduate Research Poster Symposium

2025 **Elizabeth Bottoms** *(UW-Madison)*, Women in Science and Engineering (WISE) learning community scholars program for first year undergraduate students in STEM fields

2023-2024 **Yuhan Shi** *(UW-Madison)*, Hilldale Undergraduate/Faculty Research Fellowship Awardee presented senior thesis work at the Undergraduate Research Poster Symposium & The Student Thesis and Research Showcase

2022 **Helen Treankler** *(UW-Madison)*, Individual Research Project presented at the Undergraduate Research Poster Symposium

Mentored Research Assistants

Gabriel Gutierrez, Lara Knight, Addison Peters, Samantha Baniel, Melis Baskaya, Greg Dekker, Kairui Gu, Josie Holman, Carter Klenko, Chaoran Li, Kyra Mathew, Alex Moews, Kien O'Loughlin, Camille Redford, Amanda Rhee, Jason Starr, Leila Tannous, Elsa Wei, Conlan Yourell, Emma Wagner, Sasha Vulakh, Elizabeth Bottoms, Anxhela Tata, Alexander (Alex) McGinty, Kayla Terhaar, Bret Spielbauer, Vivian Wang, Meredith Lindman, Jenna Bergmanis, Callin Dai, Manmeet Dang, Natalia De Emanuel, Zachary Fredrich, Steven Ge, Angelina Godinez, Jasmine Horrach, Ethan Hsieh, Rainy Jin, Levi Kasten, Cassie Liu, Zheyuan Liu, Vy Lu, Artem Lyu, Jacqueline Olen, Teagan Raffenbeul, Delaney Sanford, Leila Schiegg, Yuhan Shi, Helen Treankler, Eliza Starr, Allison Swid, Katelyn Van Buren, Lauren Verhulst, Caroline Wang, Emily Weiss, Alexander Wilcox, Meghan Wokurka, Yining Zhao, Li Zhan, Siyuan Yin, Xiyu Zhang

Research Experience

2021–	**Graduate Student Research Assistant**: - Learning and Transfer Lab, University of Wisconsin-Madison, (Advisor: Professor C. Shawn Green)
2020–	**Graduate Student Research Associate**: - Brain and Learning Lab, UNIGE, Faculty of Science and Education, Geneva, Switzerland, (Advisor: Professor Daphné Bavelier)
2019-2021	**Lab Manager**: - Human Robot Interaction Lab, Computational Media Department, University of California-Santa Cruz, (Advisor: Professor Leila Takayama)
2019-2021	**Undergraduate Research Assistant**: - Human Robot Interaction Lab, Computational Media Department, University of California-Santa Cruz, (Advisor: Professor Leila Takayama)
2019	**Undergraduate Research Assistant**: - High Level Perception Lab, Psychology Department, University of California-Santa Cruz, (Advisor: Professor Nicolas Davidenko)
2018-2019	**Undergraduate Research Assistant**: - Early Social Innovation Lab, Psychology Department, University of California-Santa Cruz, (Advisor: Professor Audun Dahl)

Publications

G Google Scholar

Journal Articles

J1. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. Dimensions of planning and their implications for planning performance in planning-based mini games. (2027). [2]

J2. Strobach, T., **Cunningham, Emma G.**, Astra, S. & Green, C. S. Commentary on "Do Musicians Have Better Short-Term Memory Than Nonmusicians? A Multi-lab Study": Where does this study guide future research endeavors? *Advances in Methods and Practices in Psychological Science* (2026).

J3. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. Evaluating planning through play: Exploring the use of mini games to assess planning abilities. *Acta Psychologica* **259,** 105252 (2025).

J4. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. Rethinking planning metrics: An analysis of common measurements of planning abilities. *Cognition* **263,** 106220 (2025).

[2] In prep

J5. Joessel, F., **Cunningham, Emma G.** & Green, C. S. No Expectancy Effect on Visual Attention Performance After a Single Session of Playing Sudoku: A Failed Replication of Tiraboschi et al. (2019). *Journal of Cognitive Enhancement*, 1–16 (2025).

J6. Pasqualotto, A., **Cunningham, Emma G.**, Holman, C., Bediou, B. & Bavelier, D. Digital tools for reading success: Meta-analyses of digital interventions. *Journal of Cognitive Enhancement* (2025).

J7. Moortgat-Pick, A., So, P., Sack, M. J., **Cunningham, Emma G.**, Hughes, B. P., Adamczyk, A., Sarabakha, A., Takayama, L. & Haddadin, S. *A-rift: Visual substitution of force feedback for a zero-cost interface in telemanipulation* in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2022), 3926–3933.

J8. Pasqualotto, A., Zanichelli, B., Marolo, M., **Cunningham, Emma G.** & Venuti, P. [The impact of computerized executive function training on children's reading outcomes: A review of literature]. *DIS-Dislessia, Discalculia & Disturbi di Attenzione* **2** (2021).

Invited Book Chapters

B1. **Cunningham, Emma G.** & Green, C. S. in *Games & Learning Handbook* (Oxford University Press, 2026). *

B2. **Cunningham, Emma G.** & Green, C. S. in *Oxford Research Encyclopedia of Communication.* (Oxford University Press, 2023).

B3. Parong, J., Holman, C., **Cunningham, Emma G.**, Green, C. S. & Bavelier, D. in *Using Cognitive and Affective Metrics in Education-Based Simulations and Games.* (Routledge, 2021).

Presentations

Invited Talks

T1. **Cunningham, Emma G.** *How Strategy Shapes Performance Under Limited and Full Planning Horizons: Evidence from a Novel Taskified Game.* Learning and Neuroplasticity Meeting (Campus Biotech, University of Geneva, Switzerland). Oct. 2026.

T2. **Cunningham, Emma G.** *Scrapping Together a Novel Game-Based Paradigm for Measuring Planning Abilities.* Play Make Learn Conference (Madison, WI, USA). July 2026.

T3. **Cunningham, Emma G.** *Curiosity as a Driving Force: A Journey into Research on Planning and Games.* Psi Chi UW-Madison Chapter Meeting (Madison, WI, USA). Mar. 2025.

T4. **Cunningham, Emma G.** *Evaluating Planning Through Play: Exploring the Use of Mini Games to Assess Planning Abilities.* Cognition and Cognitive Neuroscience Area Group Proseminar (Department of Psychology Madison, WI, USA). Feb. 2025.

*In press

T5. **Cunningham, Emma G.** *Components of Planning Explored Through Mini Games.* Brain and Learning Lab Group Annual Retreat (Campus Biotech, University of Geneva, Switzerland). Oct. 2024.

T6. **Cunningham, Emma G.** *Didn't Plan for That: Exploring Correlations Between Common Measurements of Planning Skills* Brain and Learning Lab Group Meeting (Campus Biotech, University of Geneva, Switzerland). May 2024.

T7. **Cunningham, Emma G.** *Individual Differences in Learning a Complex Visuo-Motor Task* Cognition and Cognitive Neuroscience Area Group Proseminar (Department of Psychology Madison, WI, USA). Oct. 2023.

T8. **Cunningham, Emma G.** *Planning Complexity: Examining Correlations Between Planning Skills and Selected Mini Games.* Annual Department of Psychology First Year Project Symposium (Department of Psychology Madison, WI, USA). Nov. 2022.

Conference Presentations

P1. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. *Dimensions of planning and their implications for planning performance in planning-based mini games.* Department of Psychology 2026 PhD Research Poster Fair (Madison, WI, USA). Apr. 2026.

P2. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. *Testing How Strategy Shapes Planning Performance Across Two Taskified Mini Game Paradigms.* Psychonomic Society Annual Conference (San Diego, CA, USA). Nov. 2026.

P3. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. *Dimensions of planning and their implications for planning performance in planning-based mini games.* Psychonomic Society Annual Conference (Denver, CO, USA). Nov. 2025.

P4. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. *Evaluating planning through play: Exploring the use of mini games to assess planning abilities.* Cognitive Science Society Annual Meeting (San Fransisco, CA, USA). July 2025.

P5. Yüksel, E. M., Anthony, L. E., **Cunningham, Emma G.**, Kasten, L. & Green, C. S. *Individual Differences in Learning a Complex Visuo-Motor Task.* Department of Psychology 2025 PhD Research Poster Fair (Madison, WI, USA). Apr. 2025.

P6. **Cunningham, Emma G.**, Anthony, L. E., Kasten, L., Yüksel, E. M. & Green, C. S. *Individual Differences in Learning a Complex Visuo-Motor Task.* Psychonomic Society Annual Conference (San Francisco, CA, USA). Nov. 2023.

P7. **Cunningham, Emma G.**, Bavelier, D. & Green, C. S. *Minimal Correlations Found Between Multidisciplinary Metrics of Planning and Planning-Related Games.* The Association for Psychological Science Annual Conference (Washington D.C., USA). May 2023.

Awards & Honors

2026 **Psychology Department Graduate Student Distinguished Service Award**
- Awarded as the Department's 2026 honoree for outstanding service efforts that significantly enhanced departmental functioning and community, recognizing work that goes well beyond normal graduate student responsibilities in research, teaching, and mentoring. ($1000)

2026 **Psychology Department Graduate Student Teaching Award**
- Chosen as the Department's 2026 recipient in recognition of exceptional teaching across introductory, methods, capstone, and graduate-level statistics courses. Honored for empowering students as confident learners and for highly thoughtful, research-informed instructional design. ($500)

2024 **ThinkSwiss Research Scholarship**
- Research award to support proposed project research completed in Geneva, Switzerland with the support of Professor Daphné Bavelier. *Secretariat for Education, Research and Innovation (SERI) & The Embassy of Switzerland in Washington D.C., USA* ($6,500)

2022-2026 **Menzies And Royalty Research Award**
- Research awards to support proposed research activities including collecting data for ongoing projects. ($500)

2022-2025 **Hertz Travel Awards**
- Travel awards to support presentation of research work at various conferences across the U.S.. (variable amounts)

2021 & 2024 **University of Wisconsin-Madison Graduate Student Fellowship**
- Year-long fellowship with accompanying stipend to support pursuance of PhD. *UW-Madison Graduate School*

2017-2021 **Schwab-Rosenhouse Memorial Scholarship**
- Scholarship awarded to high school students with excellent academic records and financial need to support pursuance of a 4 year college degree ($12,000: $3,000 yearly)

2019 **Walsh Family Scholarship at UC Santa Cruz**
- Scholarship award for excellence in academics and research at the University of California-Santa Cruz ($5,000)

2018 **Undergraduate Research College Scholars Program UCSC**
- Merit-based honors program for high achieving undergraduates interested in pursuing research.

Service

Committee & Departmental Service

2025-2026 **Graduate Student Organizer** for Women in Science and Engineering scholars mentorship program

- Responsible for connecting first year undergraduate STEM students to graduate student mentors in relevant fields

2021-2026 **Chair** of Psychology Department Graduate Student Social Events Committee
- Responsible for conceptualizing, organizing, and hosting more than 30 events to promote graduate student well-being, inter-cohort, and inter-area group connection and collaboration. Responsible for organizing and soliciting funds and raised over $1500 for the benefit of graduate students.

2024-2025 **Graduate Student Representative** to the Department Faculty Meetings
- Elected by fellow graduate students to represent the departmental graduate student body's interests to the faculty, and record and disseminate detailed notes back to graduate students about important departmental information.

2021-2024 **Member** of Psychology Department Colloquium Committee
- Responsible for deciding upon, inviting, and hosting visiting faculty presenters for regular departmental research colloquia

Ad-Hoc Journal Reviewer

Scientific Reports [Springer-Nature]
Journal of Cognitive Enhancement [Springer-Nature]
Journal of Intelligence

Tools & Software

R Packages

EGCfuncs: R Package with General Analysis and Visualization Functions. Specific function for obtaining Bayes t-test cut off values for correlational data sets GitHub).

Other Experience

2020 **UX Project Lead:**
- The Silicon Valley Social Innovation Lab @ Open Minds, Silicon Valley, Menlo Park, CA. (PI: Matthew Jacobs)

Last updated: November 2, 2026